

Given the above, our objective is to first check if the serialisation of 'student_name' and 'roll_no' is valid, followed by checking for the validity of the contents of 'detailed_data' by comparing with the schema. We can do either of the following: manually write the schema or use JSON models. The latter is recommended since it auto-generates the schema, but to get an idea of what the schema looks like, we will go through both methods.

Methods of schema validation

Writing the schema manually

To map our schema onto the target data that we want to enter in the JSONField, consider Figure 4.



```
POST http://127.0.0.1:8000/student/
JSON Auth Query Header 1 Docs
1 {
2   "student_name": "Kamil",
3   "roll_no": 85,
4   "detailed_data": {
5     "date_of_joining": "01-05-2017",
6     "mother_name": "Alice",
7     "father_name": "Matthew",
8     "teacher_information": {
9       "name": "Ms. Anne",
10      "date_of_birth": "09-10-1978",
11      "department": "102",
12      "joining_date": "05-07-2007"
13    },
14    "semester_1_subjects": [
15      {
16        "subject_name": "Maths 1",
17        "professor": "Mr. Adam",
18        "score": 75
19      },
20      {
21        "subject_name": "English 1",
22        "professor": "Ms. Evelyn",
23        "score": 80
24      }
25    ],
26    "semester_2_subjects": [
27      {
28        "subject_name": "Maths 2",
29        "professor": "Mr. Ram",
30        "score": 75
31      },
32      {
33        "subject_name": "English 2",
34        "professor": "Mr. Peter",
35        "score": 80
36      }
37    ]
38  }
39 }
```

Figure 4: Example data for JSONField

The schema suitable to fit the type of data is as follows:

```
{
  "type": "object",
  "properties": {
    "date_of_joining": {
      "type": "string",
      "pattern": "^[0-9]{2}-[0-9]{2}-[0-9]{4}$",
      "mother_name": {
        "type": "string"
      },
      "father_name": {
        "type": "string"
      },
      "teacher_information": {
        "type": "object",
        "properties": {
          "name": {
            "type": "string"
          },
          "date_of_birth": {
            "type": "string",
            "pattern": "^[0-9]{2}-[0-9]{2}-[0-9]{4}$",
            "department": {
              "type": "string"
            },
            "joining_date": {
              "type": "string",
              "pattern": "^[0-9]{2}-[0-9]{2}-[0-9]{4}$"
            },
            "required": ["name", "date_of_birth", "department", "joining_date"],
            "additionalProperties": false
          },
          "semester_1_subjects": {
            "type": "array",
            "items": {
              "type": "object",
              "properties": {
                "subject_name": {
                  "type": "string"
                },
                "professor": {
                  "type": "string"
                },
                "score": {
                  "type": "integer"
                }
              },
              "required": ["subject_name", "professor", "score"],
              "additionalProperties": false
            },
            "semester_2_subjects": {
              "type": "array",
              "items": {
                "type": "object",
                "properties": {
                  "subject_name": {
                    "type": "string"
                  },
                  "professor": {
                    "type": "string"
                  },
                  "score": {
                    "type": "integer"
                  }
                },
                "required": ["subject_name", "professor", "score"],
                "additionalProperties": false
              }
            }
          },
          "required": ["date_of_joining", "mother_
```